

**Key Points:**

- High resolution vertical velocities from DopplerScatt were combined with satellite ocean color to directly estimate net physical flux of particulate organic carbon (POC)
- Submesoscale dynamics made important contributions to POC flux below the mixed layer at a seasonally re-occurring upwelling front
- This method can be expanded to other regions to further understand the influence of submesoscale-driven flux on the marine carbon cycle

**Supporting Information:**

Supporting Information may be found in the online version of this article.

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## Estimating Submesoscale-Driven Particulate Organic Carbon Flux From Remote Sensing at a Dynamic Upwelling Front

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**Abstract** Submesoscale (1–10 km) vertical velocities at fronts enhance the exchange of particulate organic carbon (POC) and heat between the surface ocean and depth. While POC and temperature can be readily assessed in situ, direct quantification of flux remains uncertain due to the observational challenges of measuring submesoscale vertical velocities in the upper ocean. Here, we provided novel estimates of submesoscale-driven POC flux by pairing airborne-derived vertical velocities (NASA DopplerScatt) with POC estimates from satellite ocean color (Sentinel-3 OLCI) in a persistent upwelling frontal structure in the central California Current System. Instantaneous, advective fluxes of POC and heat reached up to  $\mathcal{O}(1,000 \text{ mg C m}^{-2} \text{ day}^{-1})$  and  $\mathcal{O}(1,000 \text{ W m}^{-2})$  at the front, both upward and downward. When integrated over sufficient spatiotemporal scales to achieve a negligible total volume flux, the turbulent flux, or net flux, can be estimated. Submesoscale vertical velocities made significant contributions to net downward POC flux ( $-44.1 \pm 23.1 \text{ mg C m}^{-2} \text{ day}^{-1}$ ) and upward heat flux ( $186.2 \pm 101.6 \text{ W m}^{-2}$ ). Spatial distributions of instantaneous fluxes, surface kinematics, and ship-based, high-resolution bio-optical and hydrographic profiles indicated that POC flux below the mixed layer was driven by frontal overturning and filament subduction. This study contributes to the understanding of how submesoscale processes modify carbon export in the highly productive California Current System. This method can be expanded to other regions to further understand the influence of submesoscale-driven flux on the marine carbon cycle.

**Plain Language Summary** Oceanic processes acting on 1–10 km spatial scales enhance the vertical transport of carbon and heat between the surface ocean and depth. The undissolved component of organic carbon in the ocean, particulate organic carbon (POC), can be estimated with satellites. However, vertical velocities are challenging to measure, and therefore direct estimates of the physical vertical flux of POC remains uncertain. In this study, we provided novel direct estimates of physical POC flux by pairing airborne-derived vertical velocities with POC estimates from satellite ocean color in a seasonally re-occurring upwelling front in the central California Current System. The net vertical fluxes of POC and heat were  $-44.1 \pm 23.1 \text{ mg C m}^{-2} \text{ day}^{-1}$  (downward) and  $186.2 \pm 101.6 \text{ W m}^{-2}$  (upward). We also characterized the processes driving flux with physical and biological data from remote sensing and a research vessel. By providing novel estimates of physical POC flux and characterizing the associated dynamics, this study contributes to the understanding of how processes at 1–10 km scales modify carbon export in the highly productive California Current System. This measurement approach can be implemented in other regions to further understand the influence of small-scale oceanic processes on the marine carbon cycle.

### 1. Introduction

Ocean carbon sequestration and the productivity of marine ecosystems rely on the vertical exchange of particulate organic carbon (POC) from the surface ocean to depth. Submesoscale vertical velocities, occurring at a Rossby number ( $\zeta/f$ ) of  $\mathcal{O}(1)$  and typical length scales of  $<10 \text{ km}$ , can facilitate the physical transport of carbon, heat, nutrients, and other properties, between the surface ocean and depth (D'Asaro et al., 2018; Freilich et al., 2024; Omand et al., 2015; Spall, 1995; Su et al., 2018; Thomas et al., 2008; Torres et al., 2025). The duration and depth of this exchange may impact ocean carbon sequestration and help offset atmospheric carbon concentrations (De

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La Rocha & Passow, 2014). Vertical advective transport of biogeochemical tracers can be intensified by a number of submesoscale processes, such as overturning circulation at fronts and density-driven instabilities (Mahadevan, 2006; Mahadevan & Tandon, 2006). The sharpening of lateral buoyancy gradients by frontogenesis induces an ageostrophic secondary circulation with downwelling on the dense side of the front and upwelling on the light side of the front (Hoskins, 1982; McWilliams, 2021). The front will sharpen until frontal arrest occurs, yielding favorable conditions for the formation of baroclinic instabilities. Instabilities can develop into dense submesoscale filaments with large positive vorticities, lateral strains, and vertical velocities (Sullivan & McWilliams, 2018; Thomas et al., 2008).

Localized, instantaneous fluxes of POC can be as high as  $\mathcal{O}(1,000 \text{ mg C m}^{-2} \text{ day}^{-1})$  downward, as diagnosed from models (Resplandy et al., 2019), but the cumulative impact of vertical velocities on net flux depends on the compensation between upwelled and downwelled particulate and dissolved organic carbon (Bellacicco et al., 2025; Le Chevère et al., 2026; Omand et al., 2015; Resplandy et al., 2019). Although the impact of submesoscale velocities on global POC flux and sequestration is uncertain, the net flux is likely much less than these large instantaneous flux events (Resplandy et al., 2019). Observationally, instantaneous flux events are often identified by localized anomalous properties such as enhancement in chlorophyll-*a* (chl-*a*), POC, and other properties (e.g., spice, oxygen), captured in ship-based profiles of paired hydrographic and bio-optical data at fronts (Chen & Schofield, 2024; Freilich et al., 2024; A. R. Johnson & Omand, 2021; Omand et al., 2015). These ubiquitous profile characteristics are fingerprints of instantaneous fluxes, but preclude a direct estimate without corresponding vertical velocity measurements or statistical approximations. Additionally, observational studies are often biased toward mesoscale processes and large instantaneous transport events due to the difficulty of resolving submesoscales over sufficient spatiotemporal domains.

The challenge of accurately measuring vertical velocities 3–4 orders of magnitude smaller than horizontal velocities is significant (Mahadevan & Tandon, 2006). Instead, vertical velocities are often inferred from more easily measured quantities. The quasigeostrophic Omega equation, for example, is commonly used to infer vertical velocities from 3D-resolved density fields (Allen et al., 2001; Hoskins et al., 1978; Pinot et al., 1996; Rudnick, 1996). Other approaches, such as surface quasigeostrophy and its variations (e.g., effective and interior surface quasigeostrophy) are used to reconstruct  $w$  from surface measurements like satellite sea surface height (Lapeyre & Klein, 2006; Wang et al., 2013). These approaches assume a small Rossby number, which does not hold true for the submesoscale, but there has been some progress in using effective SQG to reconstruct vertical velocities and heat fluxes from the Surface Water and Ocean Topography Mission (SWOT) at 20–100 km scales (Carli et al., 2024, 2025).

Vertical velocity can also be inferred from horizontal divergence via continuity, which can be achieved using total current measured by drifters (Tarry et al., 2021, 2022) or remote sensing (Chelton, 2023; Lang et al., 2025; Lenain et al., 2023). Airborne sensors are enabling exciting opportunities to capture high-resolution horizontal current gradients. For example, NASA's airborne Ka-band Doppler scatterometer, DopplerScatt, measures high resolution (0.2 km spatial resolution) spatial maps of surface currents from which vertical velocities ( $w$ ) can be estimated from horizontal divergence (Rodríguez et al., 2018; Torres et al., 2025).

Instantaneous, advective fluxes of POC can be parameterized by the product of the  $w$  anomaly and POC anomaly,  $w' \text{POC}'$ , where anomalies are defined as fluctuations from spatial or temporal averages (e.g., Omand et al., 2015). The ensemble average of the fluctuating instantaneous, advective fluxes,  $\overline{w' \text{POC}'}$ , is known as the turbulent or eddy flux (Liu et al., 2023; Montgomery, 1948; Moum, 1996; Pond et al., 1971; Priestley & Swinbank, 1947; Swinbank, 1951). Turbulent flux is a statistical description of advection assessed at scales which no longer resolve individual advective motions (Liu et al., 2023; Taylor, 1915). The turbulent, or net, fluxes of POC and heat ( $Q$ ) will be represented by  $F_{\text{POC}}$  and  $F_Q$  and calculated by taking the spatiotemporal averages of  $w' \text{POC}'$  ( $F_{\text{POC}}^*$ ) and  $w' Q'$  ( $F_Q^*$ ). Selecting the appropriate time and space scales that are fine enough to capture these episodic events, but broad enough to ensure that there are no unbalanced vertical velocities, is a crucial part of the analysis. Averaging must be extensive enough to ensure no net volume flux ( $\bar{w} = 0$ ); otherwise, the unbalanced, instantaneous advective fluxes are likely to overshadow the relatively small turbulent flux.

This study aims to quantify the net POC flux at a restratifying upwelling front by pairing satellite ocean color ( $\sim 0.3$  km spatial resolution) with DopplerScatt currents ( $\sim 0.2$  km spatial resolution). We also estimated the net vertical heat flux for comparison (Torres et al., 2025). In Eastern Boundary Upwelling Current Systems, such as

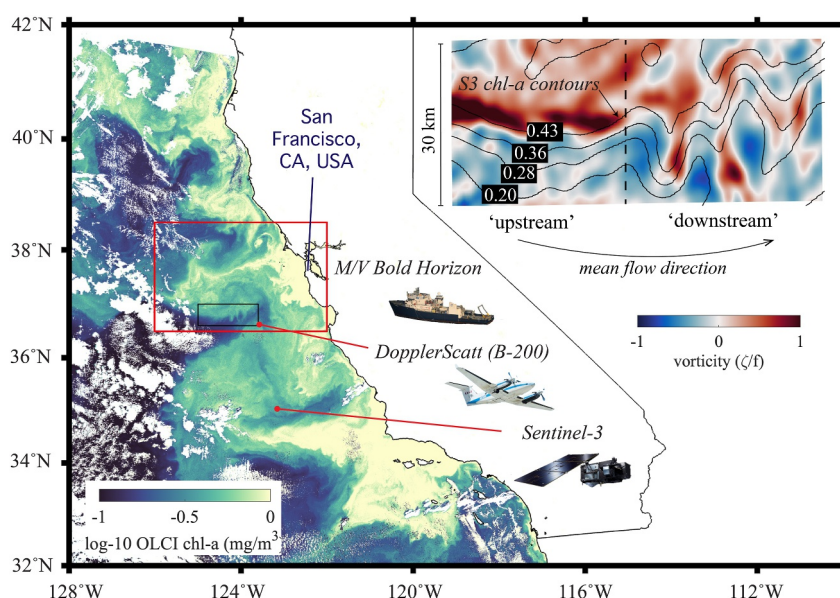
the California Current System, positive buoyancy flux from frontal restratification is associated with positive vertical heat flux (Colas et al., 2013; L. Johnson, Lee, D'Asaro, Thomas, & Shcherbina, 2020; L. Johnson, Lee, D'Asaro, Wenegrat, & Thomas, 2020; Torres et al., 2025). Here, we found that positive vertical instantaneous fluxes of heat, estimated with satellite sea surface temperature (SST, 1 km spatial resolution), were correlated with negative instantaneous fluxes of POC ( $R^2 = 46\%$ ), arising from the correlations between (cold) temperature and (high) POC anomalies typically observed in upwelling systems. We estimated a net POC flux of  $-44.1 \pm 23.1 \text{ mg C m}^{-2} \text{ day}^{-1}$  (downward) and a coincident net vertical heat flux of  $186.2 \pm 101.6 \text{ W m}^{-2}$  (upward). We find that our study front is a consistent feature of a persistent mesoscale structure that re-occurs seasonally—likely shaped by the bathymetry of the coastline and modulated by the seasonality of upwelling and transport (Centurioni et al., 2008; Checkley & Barth, 2009; Giddings et al., 2022). Our findings contribute to the understanding of how submesoscale features evolve in time and space, and ultimately contribute to carbon export in the highly productive central California Current System.

## 2. Overview of Study Region and Its (Sub)mesoscale Variability

The data in this study were collected between 23–27 October 2022, during the second field campaign of the Submesoscale Ocean Dynamics Experiment (S-MODE). S-MODE was a NASA Earth Venture Suborbital-3 (EVS-3) mission that tested the hypothesis that submesoscale eddies and fronts drive significant vertical exchanges of climatic and biologically-relevant properties in the upper ocean, such as heat and organic carbon (Farrar et al., 2025). The multi-platform, multi-scale mission consisted of three aircraft with various surface ocean-observing payloads, a research vessel (M/V Bold Horizon), autonomous vehicles (e.g., Wavegliders, Saildrones, Lagrangian floats, drifters, gliders), and satellite observations. The S-MODE study region extended about 200 km off the coast of San Francisco in the central California Current System (CCS; Figure 1).

The observations of interest in this study were surface and subsurface measurements of paired physical and biological variables from the M/V Bold Horizon, SST and chlorophyll-a measured by sensors onboard Sentinel-3, and high resolution surface currents from the airborne instrument DopplerScatt (Figure 1). Unlike satellite altimetry, DopplerScatt measures ageostrophic currents at the fine spatial scales (0.2 km) necessary for resolving submesoscale flow. An example of DopplerScatt-derived vorticity at the front of interest, occurring close in time to a clear Sentinel-3 chl-a image, is shown in Figure 1. Relative vorticity, which approaches the Coriolis frequency at submesoscales, is well aligned with chl-a variability as shown by the alignment of chl-a contours (black) overlain on vorticity in the inset map in Figure 1. The front was characterized by two dynamical regimes on the days of DopplerScatt overpasses (October 26th and 27th): upstream and downstream. Although separated in space, the domains represent different stages of frontal evolution based on their inferred stability. The regimes were demarcated based on the sharpness of the front exhibited in chl-a, SST, and vorticity upstream and filamentous structures prevalent in the downstream domain (Figure 1). The remainder of this section will describe the mesoscale processes which set up the front and give an overview of the front's spatiotemporal evolution.

The California Current System, like other Eastern Boundary Upwelling Systems, contributes disproportionately to primary productivity in the ocean by upwelling cold, nutrient rich waters that fuel surface phytoplankton growth and high POC concentrations. The cold, highly productive water is subsequently transported offshore through horizontal stirring by the geostrophic strain field resulting in subduction and mixing with warmer, oligotrophic waters further offshore. The intensity of horizontal stirring impacts mesoscale chlorophyll-a variability and the formation of upwelling fronts, which can become unstable and form meanders and instabilities (Capet et al., 2008b; Lehahn et al., 2007; Lévy et al., 2018; Giddings et al., 2022; Rossi et al., 2009). Stirring intensity also varies spatially and temporally in the CCS due to correlations with seasonal upwelling and internal climate variability (Giddings et al., 2022). The central CCS off the coast of San Francisco typically has intensified stirring relative to the rest of the CCS, and October hosts particularly high mesoscale activity on average due to high upwelling intensity (Giddings et al., 2022; Jorgensen et al., 2024). The persistence and seasonal context of our observations are addressed in the Discussion section, but we start our results with an overview of the mesoscale chl-a distributions during the focus period (Oct. 23–27), with geostrophic current streamlines overlain (Figures 2a–2d). Recently upwelled water, denoted by high chl-a concentrations (yellow and green in Figures 2a–2d), was actively being injected and stirred offshore, running into a warm, low chl-a water mass (typical of offshore water types) and advecting laterally. The portion of the jet that was re-directed to the southeast set up the density front of interest denoted by the black box in Figures 2a–2d and plotted in Figures 2e–2l.



**Figure 1.** Log-10 chl-a from the Sentinel-3 Ocean and Land Color Instrument (OLCI, 0.3 km spatial resolution) in the study region off the coast of California, USA. Red box denotes the mesoscale chl-a feature, which sets up the front of interest denoted by the smaller black box. Photos of the measurement platforms used in this study include the M/V Bold Horizon, DopplerScatt onboard the King Air B-200, and Sentinel-3. Vorticity derived from DopplerScatt surface currents with Sentinel-3 chl-a contour overlain in black (inset). Contour labels are associated with units [ $\text{mg m}^{-3}$ ]. The upstream domain represents an earlier stage of frontal evolution characterized by a sharper front and the downstream domain represents a more unstable stage of frontal evolution characterized by the development of instabilities and filaments. Credits for the measurement platform photos, in order: Alex Wineteer, NASA, European Space Agency.

At the time of the satellite, shipboard, and aircraft (DopplerScatt) surveys analyzed here, the mesoscale front of interest had developed meanders as shown in chl-a and SST on October 23rd and 24th (Figures 2e, 2f, 2i, and 2j). The next snapshot on October 26th, which coincided with DopplerScatt measurements, revealed a sharper front upstream, where the DopplerScatt-derived streamlines showed confluent flow toward the front (longitudes:  $-124.3$  to  $-125$ ; Figures 2g and 2k), and the formation of distinct submesoscale filaments downstream, where DopplerScatt streamlines began to spread out (longitudes:  $-123.6$  to  $-124.3$ ; Figures 2g and 2k). The vertical structure and surface kinematics of one of these filaments (SF1 for “submesoscale filament 1”) will be further explored in Section 4.2 and Text S3, Figure S15 in Supporting Information S1. On October 27th, SF1 is largely gone and other filamentous structures are more diffuse (Figures 2h and 2l). This study seeks to quantify the vertical flux of POC and heat associated with the evolution of this front, the vertical extent of the subsurface subduction that influences the sequestration impact, and the vorticity-strain relationship that reveals dynamical underpinnings.

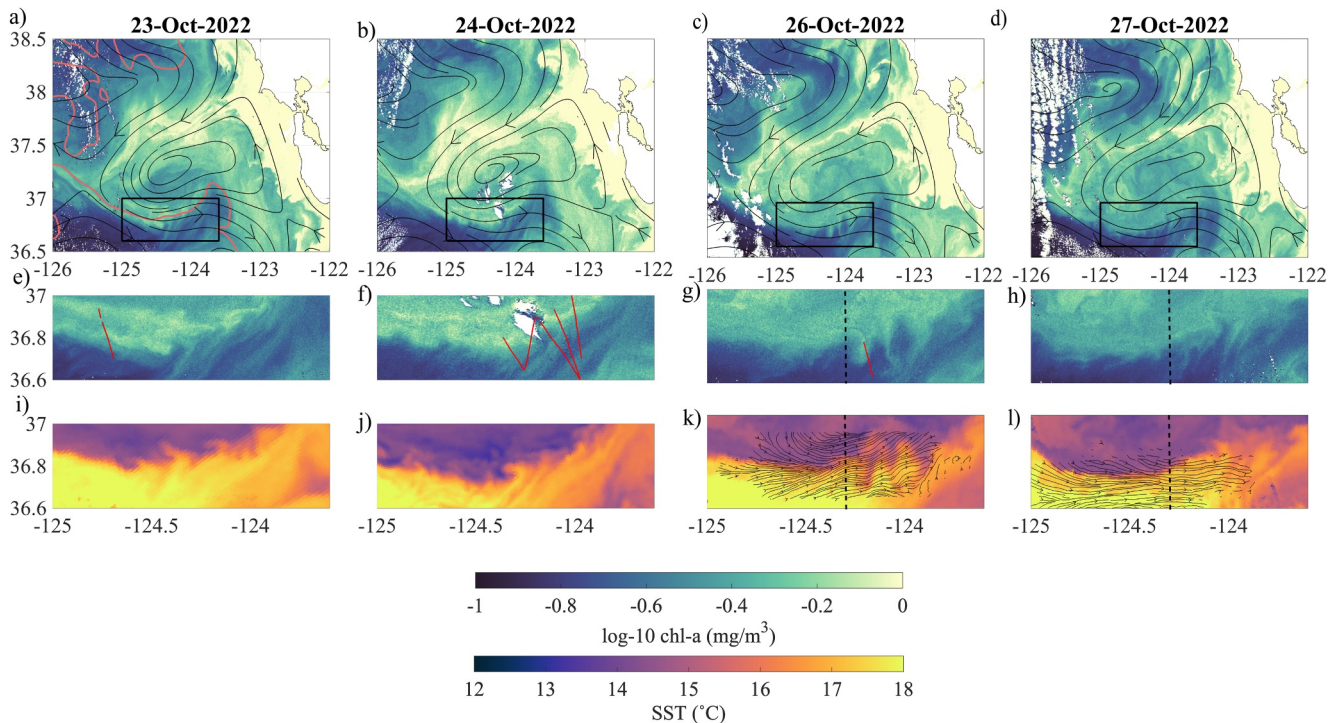
### 3. Methods

#### 3.1. Overview

Data used in this study were collected with instruments onboard the M/V Bold Horizon, a King Air B-200 aircraft, and Sentinel-3 satellites. Torres et al. (2025) combined airborne ocean velocities with concurrent airborne SST observations to estimate the submesoscale-driven vertical heat flux at the front of interest, albeit over a different temporal range. Here, we took a similar approach to estimate the submesoscale-driven POC flux and associated vertical heat flux using co-located Sentinel-3 ocean color and SST taken close in time to airborne overpasses ( $<30$  min outside of airborne sampling window). Concurrent analysis of the vertical heat and POC fluxes allows us to highlight the strong connections between heat flux and submesoscale-driven carbon export in upwelling systems as described in Section 5.1.

We derived POC concentrations from a linear relationship between Sentinel-3 chl-a and a bio-optical POC proxy measured by the M/V Bold Horizon's underway flow-through system. Next, we calculated  $F_{\text{POC}}$  and  $F_Q$  at varying





**Figure 2.** Sentinel-3 OLCI chl-a on 23, 24, 26, and 27 October 2022 with COPERNICUS-GLOBCURRENT geostrophic streamlines overlain in black (a–d).  $16^\circ$  contour, overlain in pink on (a), indicates that the high chl-a plume extending offshore is associated with colder sea surface temperature (SST) than the surrounding low chl-a waters. Front of interest denoted by black boxes (a–d). Chl-a plotted for front of interest (e–h) with EcoCTD transects (red lines) overlain (e–g). Five transects surveyed the front between October 23–25 (red lines, e, f) and one transect surveyed the submesoscale filament on October 26 (red line, g). Sentinel-3 SLSTR SST 1 km spatial resolution plotted for frontal region of interest (i–l) with same day DopplerScatt total current streamlines overlain in black (k, l). Vertical dashed line denotes demarcation between upstream and downstream (g, h; k, l).

smoothing scales (1–15 km) to assess the scale dependence of flux estimation and select the optimal smoothing scale. We then investigated the spatial distributions of  $F_{\text{POC}}^*$  and  $F_Q^*$  in a frontal coordinate system. Lastly, we conducted an analysis of the vertical physical and biological structures of the front and filament of interest, SF1, using ship-based hydrographic, bio-optical, and velocity data from an underway profiler and an Acoustic Doppler Current Profiler (ADCP).

### 3.2. Data Collection and Processing

#### 3.2.1. Airborne Data

Airborne measurements of surface currents and SST were measured by DopplerScatt (NASA) and MOSES (UCLA, Ifremer) onboard a King Air B200 aircraft. DopplerScatt is a Ka-Band (35.75 GHz) Doppler Scatterometer that measures concurrent winds and 4 m depth surface velocities at a 0.2 km spatial resolution over a 24 km swath (Rodríguez et al., 2018, 2023). Vorticity, strain, and divergence were derived from DopplerScatt Level 2 Version 2 surface currents to characterize the flow structure and estimate vertical velocities (Rodríguez et al., 2023). DopplerScatt data were cross-calibrated against in situ measurement platforms (Acoustic Doppler Current Profilers, surface drifters) and optical imagery (DoppVis instrument; Lenain et al., 2023) during the S-MODE campaign. Reciprocal overpasses, imaging the surface at varying azimuth directions, reduced measurement noise and improved accuracy of current retrievals (Farrar et al., 2025; Rodríguez et al., 2018).

First, zonal and meridional velocities ( $u$ ,  $v$ ) with errors greater than 0.05 m/s were removed.  $u$  and  $v$  were smoothed with a Gaussian filter of varying smoothing scales (1–15 km). Vorticity ( $\zeta = v_x - u_y$ ), strain ( $\sigma = [(u_x - v_y)^2 + (v_x + u_y)^2]^{1/2}$ ), and divergence ( $\delta = u_x + v_y$ ) were calculated using smoothed  $u$  and  $v$ . Data along the edges of the swath were manually removed to avoid edge effects arising from sensor errors and spatial gradient calculations. DopplerScatt longitudes and latitudes were converted to Universal Transverse Mercator

(UTM) coordinates. Easting coordinates were normalized by subtracting  $3 \times 10^5$  m and northing coordinates were normalized by subtracting  $4 \times 10^6$  m. Both were then converted to kilometers.

MOSES, or the Multi-Scale Observation System of the Ocean Surface, was used for comparison to Sentinel-3 SLSTR SST. MOSES comprises of a FLIR A6751 SLS longwave infrared (LWIR) camera with a  $\sim 10$  m spatial resolution, 5.4 km swath width, and 0.1 K accuracy (Farrar et al., 2025; Molemaker & Hypolite, 2022; Torres et al., 2025). MOSES is known to have a cold bias and is corrected with an average offset based on satellite or in situ data (8.5°C average offset from SLSTR).

### 3.2.2. Satellite Data

Sentinel-3 satellite data was paired with DopplerScatt velocities to estimate the vertical flux of POC and heat. Level-2 Sentinel-3 OLCI chlorophyll-a (chl-a) and SLSTR SST (Full Resolution) were collected from the European Space Agency's Copernicus Data Browser. Sentinel-3 provides ocean color and SST from the same platform and at suitable spatial resolutions to resolve submesoscale features. Chl-a has a 0.3 km spatial resolution and SST has a 1 km spatial resolution. Sentinel-3 chl-a and SST were regridded to DopplerScatt grids (0.2 km pixel size) and smoothed with a Gaussian filter to match the smoothing scales of Dopplerscatt velocities. Sentinel-3 chl-a was converted to POC using calibrated flow-through bio-optical data (Figure S1 in Supporting Information S1).

### 3.2.3. Coupling Airborne and Satellite Data

Three B200 research flights surveyed the front of interest on days with clear Sentinel-3 ocean color and SST imagery. These three flights and their paired Sentinel-3 ocean color data will be referred to as Oct. 26M, Oct. 26A, and Oct. 27M. Two flights took place on 26 October at 09:23–12:38 Pacific Daylight Time (PDT) (Oct. 26M) and at 14:59–18:24 PDT (Oct. 26A). One flight took place on 27 October at 11:54–15:05 PDT (Oct. 27M). Sentinel-3 overpasses collected data between 11:20 and 11:23 PDT on the 26th, which coincided with Oct. 26M and between 11:33 and 11:36 PDT on the 27th, which coincided with the Oct. 27A.

To utilize DopplerScatt data from Oct. 26A, ocean color and SST fields from Oct. 26M were advected forward in time 6.08 hr to yield predicted spatial match-ups close in time to the B200 overpass. For each grid cell, the original position ( $x_0, y_0$ ) and corresponding zonal and meridional velocities ( $u, v$ ) from Oct. 26M were used to estimate a final position ( $x, y$ ) during Oct. 26A. Final positions were given by  $x = x_0 + D \sin(\theta)$  and  $y = y_0 + D \cos(\theta)$ , where  $D = 6.08 \sqrt{u^2 + v^2}$  and  $\theta = \tan^{-1}(v/u)$ . The new fields of chl-a and SST were then regridded to the 0.2 km resolution grid. To assess the validity of the advection scheme, resulting SST fields were visually compared to SST measured by MOSES during the B-200 afternoon flight (Oct. 26A). Spatial patterns agree very well between advected Sentinel-3 SST and MOSES SST (Figure S2 in Supporting Information S1) suggesting that a linear advection over the 6.08 hr period was a reasonable approach. Sentinel-3 POC and SST and DopplerScatt vorticity, strain, and divergence fields for the resulting Oct. 26A (in addition to 26 and 27M) are plotted in Figure S3 in Supporting Information S1.

### 3.2.4. M/V Bold Horizon

The M/V Bold Horizon surveyed the front between October 23rd and 27th using an underway flow-through system and an EcoCTD vertical profiler. Continuous flow-through bio-optical measurements, taken at the near-surface, were used to calibrate Sentinel-3 POC. Flow-through proxies of chl-a and POC were derived from absorption line height at 676 nm and beam attenuation at 657 nm measured by an ac-s Spectral Absorption and Attenuation Sensor (Sea-Bird Scientific; Bishop, 1999; Roesler & Barnard, 2013). Flow-through proxies were calibrated with bottle samples analyzed for chl-a ( $N = 154$  samples) with a benchtop Turner A10 fluorometer and POC ( $N = 147$  samples) analyzed with a Costech 4010 Elemental Analyzer (Lang et al., 2023). Flow-through bio-optical measurements had significant correlations with bottle samples of chl-a and POC (chl-a:  $R^2 = 75.9\%$ ; POC:  $R^2 = 64.4\%$ ; Lang et al., 2023).

The vertical hydrographic and bio-optical structure of the front was assessed using a vessel-towed EcoCTD. The EcoCTD is an underway profiler capturing simultaneous, high resolution hydrographic and biological data (Dever et al., 2020). Temperature and salinity were measured by an RBR Concerto, and chlorophyll-fluorescence (chl-fl) and backscatter at 700 nm were measured by a WetLabs BB2F ECOPuck (Sea-Bird Scientific). Temperature and salinity were calibrated to match two consistent Conductivity, Temperature, Depth (CTD) sensors on a CTD-

Rosette. Chl-fl was calibrated with bottle samples of chl-a, taken at various depths, from a CTD-Rosette. A full description of calibration procedures for in situ bio-optics can be found in the S-MODE Bio-optical Calibrations Report, IOP-1 (Lang et al., 2023). Chl-fl is often used as a proxy for phytoplankton biomass, but is distinct from chl-a because it is significantly affected by non-photochemical quenching (NPQ). NPQ describes a physiological response from phytoplankton to high light conditions (Xing et al., 2018). A POC proxy was based on backscatter at ~700 nm and calibrated with bottle samples of POC taken onboard the CTD-Rosette. EcoCTD bio-optical measurements had significant correlations with bottle samples of chl-a and POC (chl-fl:  $R^2 = 75.4\%$ ; POC:  $R^2 = 49.9\%$ ). The S-MODE team collected 393 vertical EcoCTD profiles during 6 transects crossing the front between October 23–25th (red tracks in Figures 2e and 2f) and 39 profiles across SF1 on October 26th (red tracks in Figure 2g). The vertical structure of the horizontal velocities was assessed using a hull-mounted Nortek Signature100 100 kHz ADCP. Zonal and meridional velocities concurrent with EcoCTD profiles were isolated. Four EcoCTD transects had suitable corresponding velocity data from the ADCP.

### 3.3. Estimating Submesoscale-Driven Vertical Particulate Organic Carbon and Heat Flux From Remote Sensing

We estimated the submesoscale-driven POC flux,  $F_{\text{POC}}$ , as the covariance of vertical velocity submesoscale anomaly ( $w'$ ) and POC submesoscale anomaly ( $\text{POC}'$ ),  $\overline{w'\text{POC}'}$  (e.g., Omand et al., 2015).  $F_{\text{POC}}$  represents the ensemble average of  $w'\text{POC}'$  ( $F_{\text{POC}}^*$ ) over the spatiotemporal domain of the three DopplerScatt flights surveying the front. The vertical velocity ( $w$ ) was inferred from DopplerScatt horizontal divergence at the surface ( $\delta$ ), which can be related to  $w$  assuming incompressible flow ( $\delta = u_x + v_y = -w_z$ ; e.g., Farrar et al., 2025; Lenain et al., 2023; Torres et al., 2025).  $\delta$  was integrated over a depth scale  $H$ , yielding  $w = H\delta$  when  $H$  is represented as positive. It may be reasonable to assume that  $H$  scales with the mixed layer depth, over which density is approximately homogeneous.  $\delta$  was assumed to be constant over  $H$ . The vertical structure of  $\delta$  and the use of  $H\delta$  to infer  $w$  were both assessed using high-resolution (200 m horizontal resolution) model output of the S-MODE study region from the Regional Ocean Modeling System (ROMS; Text S1; Figures S4 and S5 in Supporting Information S1).

We performed a Reynolds decomposition of the POC flux, where the submesoscale-driven, or eddy flux, components are parameterized as  $\overline{w'\text{POC}'}$  ( $F_{\text{POC}}$ ):

$$\overline{w'\text{POC}'} = \overline{\delta'H\text{POC}'} + \overline{\delta'H'\text{POC}'} \quad (1)$$

$\overline{\delta'H\text{POC}'}$  represents the net submesoscale flux due to the covariance between surface divergence and POC concentrations, given a constant mixed layer depth.  $\overline{\delta'H'\text{POC}'}$  represents the submesoscale flux driven by mixed layer restratification. The derivation of Equation 1 from a Reynolds decomposition of the flux can be found in Text S2 in Supporting Information S1.  $\overline{\delta'H'\text{POC}'}$  is more difficult to assess because we cannot estimate a spatially-resolved  $H$  from remote sensing. We assessed the relative contributions of  $\overline{\delta'H'\text{POC}'}$  in the upstream region of the front, where we could estimate mixed layer depth variability as a function of distance to the front using EcoCTD data. Our analysis showed that  $\overline{\delta'H'\text{POC}'}$  was small compared to  $\overline{\delta'H\text{POC}'}$  (see Section 4.1); and therefore, we estimate  $\overline{w'\text{POC}'}$  as:

$$\overline{w'\text{POC}'} \approx \overline{\delta'H\text{POC}'} \quad (2)$$

$\overline{\delta'H\text{POC}'}$  units of  $\mu\text{mol m L}^{-1} \text{ s}^{-1}$  were converted to  $\text{mg C m}^{-2} \text{ d}^{-1}$  using a reference density,  $\rho_0$ , of  $1,025 \text{ kg m}^{-3}$ .

Similarly, we calculated vertical heat flux,  $F_Q$ , with Sentinel-3 SST to compare to POC flux:

$$\overline{w'\text{POC}'} \approx \overline{\delta'HQ'} = \rho_0 C_p \overline{\delta'HSST'} \quad (3)$$

where  $C_p$  is the specific heat capacity of water,  $4,007 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$ , and  $\text{SST}'$  represents the SST anomaly (Torres et al., 2025). As with  $\overline{\delta'H\text{POC}'}$ ,  $\rho_0 C_p \overline{\delta'HSST'}$  was small relative to  $\rho_0 C_p \overline{\delta'HSST'}$ . Positive flux denotes a net upward flux of POC or heat, and negative flux denotes a net downward flux of POC or heat.

POC', SST', and  $\delta'$  were defined by subtracting the mesoscale fields from the total fields. First, POC, SST,  $u$ , and  $v$  were smoothed with a 3 km Gaussian filter. Smoothed  $u$  and  $v$  were used to calculate  $\delta$ . The mesoscale fields ( $\bar{\delta}$ ,  $\bar{\text{POC}}$ ,  $\bar{\text{SST}}$ ) were calculated for Oct. 26M, Oct. 26A, and Oct. 27M using linear least squares fits between the  $y$ -coordinates and the variable of interest to remove the North-South variability introduced by the mesoscale eddy (Figure S6 in Supporting Information S1). At 3 km smoothing scales, the submesoscale fields (POC', SST', and  $\delta'$ ) described 21.3%, 21.7%, and 13.0% of the variability in total POC, 23.0%, 26.7%, and 18.2% of the variability in total SST, and 98.6%, 100%, and 98.6% of the variability in total  $\delta$ . Distributions of  $w'$  ( $\delta'H$ ) are nearly symmetric around 0 (skewness = 0.10; Figure S7 in Supporting Information S1).

$\bar{H}$  was calculated by taking a weighted average of mixed layer depths from the EcoCTD, which was found to be 24.01 m (standard deviation = 8.1 m). Shallow mixed layer depths on the light side of the front were over-represented in the data and were weighted by half before calculating the average (Figure S8 in Supporting Information S1).  $H'$  represents the deviation of the mixed layer from its average value  $\bar{H}$ .

The means and standard deviations of the ensemble averages,  $F_{\text{POC}}^*$  and  $F_Q^*$ , at different smoothing scales were used to characterize the scale dependence of flux estimation and select the optimal smoothing scale (Figure S9 in Supporting Information S1).  $F_{\text{POC}}$ ,  $F_Q$ ,  $F_{\text{POC}}^*$  and  $F_Q^*$  reported hereafter are associated with 3 km smoothing scales. Standard errors of the mean were quantified for  $F_{\text{POC}}$  and  $F_Q$  and the statistical significance of the correlation between  $w'$  and POC' was tested using a  $t$ -test. The degrees of freedom were assessed by subtracting 1 from the number of closed contours given by the isolines of  $w'\text{POC}' \sim 0$  (Figure S10 in Supporting Information S1). Here, we assumed that each patch of upward or downward transport represented an independent observation of  $w'\text{POC}'$ . Error quantification and the  $t$ -test for statistical significance were repeated for heat flux.

$F_{\text{POC}}^*$  and  $F_Q^*$  were analyzed in an across-frontal reference frame. Frontal boundaries were found for remote sensing data with a temperature threshold ( $T = 16.05^\circ\text{C}$ ) which visually aligned with the strongest meridional SST gradients in the western domain. Then, the front was further smoothed with a 20 km moving mean, which captured the curvature of the front but smoothed out the eddying regions downstream. Next, the minimum distance to the frontal boundary was calculated for each grid cell (Figure S11 in Supporting Information S1). Data was binned into 5 km bins. The across-frontal reference frame defines the front at  $x = 0$  km, and figures plotted in this reference frame have their  $x$ -axes labeled "distance across front (km)." Distances to the south of the front (less dense side) are negative ( $x < 0$  km) and distances to the north of the front (denser side) are positive ( $x > 0$  km).

### 3.4. Analysis of Vertical Frontal Structure

The vertical structures of chl-*fl*, POC, temperature, salinity, lateral and vertical buoyancy gradients, zonal and meridional velocities, and mixed layer depth were characterized at the front and SF1 using EcoCTD and ADCP data. We calculated the lateral ( $M^2 = \frac{-g}{\rho} \frac{\partial \rho}{\partial y}$ ) and vertical ( $N^2 = \frac{-g}{\rho} \frac{\partial \rho}{\partial z}$ ) buoyancy gradients from EcoCTD profiles. ADCP zonal and meridional velocities ( $u$ ,  $v$ ) were used to estimate along and across front velocities due to the nearly horizontal orientation of the front. Mixed layer depths were calculated with a density threshold  $\Delta\rho$  of  $0.125 \text{ kg/m}^3$  from the near surface density value,  $\rho_{\text{ref}}$ .  $\rho_{\text{ref}}$  was found by taking the median of the first two pressures closest to the surface (8–9 m) for each profile.

We calculated an "average mesoscale front transect" to characterize the background mesoscale physical and biological frontal structure. First, we binned and averaged EcoCTD data and corresponding ship-based ADCP data in an across-frontal reference frame. To do this, we interpolated between the locations of the largest density gradients for each of the 6 EcoCTD frontal passes (red EcoCTD profiles in Figures 2e and 2f). Distance from the interpolated frontal boundary were calculated for each profile. We then binned the profiles into 500 m horizontal bins ( $\sim$  horizontal resolution of the EcoCTD) defined relative to the front at  $x = 0$  km, and calculated the average vertical profile of each bin. We also characterized the vertical structure of SF1 in an across-frontal reference frame (red EcoCTD profiles in Figure 2g). We found the location of the highest density gradient in the EcoCTD transect to define the edge of the filament. The distance from the edge of the filament was calculated for each profile to define the across-frontal reference frame, and data was binned into 500 m horizontal bins and averaged.



## 4. Results

### 4.1. Vertical Fluxes of Particulate Organic Carbon and Heat

The net vertical submesoscale-driven POC flux,  $F_{\text{POC}}$ , was  $-44.1 \pm 23.1 \text{ mg C m}^{-2} \text{ d}^{-1}$  (downward) and the net heat flux,  $F_Q$ , was  $186.2 \pm 101.6 \text{ W m}^{-2}$  (upward). Correlations between  $\text{POC}'$  ( $\text{SST}'$ ) and  $w'$  were weak ( $R = -0.16$ ;  $R = 0.15$ ), but standard errors indicated that net flux was significantly different from 0. The correlations were statistically significant at the 95% and 90% confidence levels (POC:  $t(144) = -1.99$ ,  $P = 0.049$ ; heat:  $t(154) = 1.72$ ,  $P = 0.07$ ; Figure S12 in Supporting Information S1).

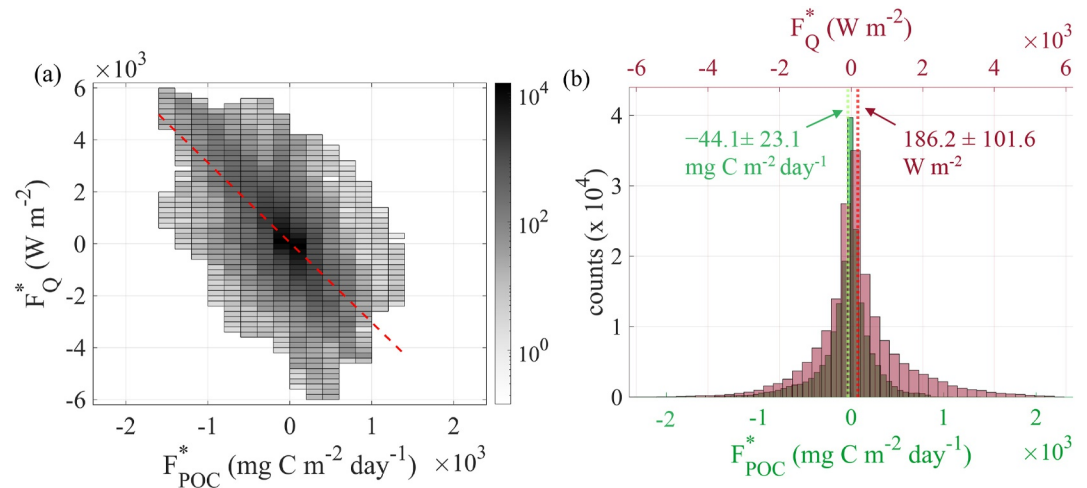
Overall,  $F_{\text{POC}}^*$  and  $F_Q^*$  were negatively correlated ( $R^2 = 46\%$ ; Figure 3a). Differences in skewness are highlighted in the dual histogram plotting distributions of  $F_{\text{POC}}^*$  and  $F_Q^*$  (Figure 3b).  $F_{\text{POC}}^*$  and  $F_Q^*$  were heavy-tailed, reaching values of  $\mathcal{O}(\pm 1,000) \text{ mg C m}^{-2} \text{ d}^{-1}$  and  $\mathcal{O}(\pm 1,000) \text{ W m}^{-2}$ , although their averages were much less.  $F_{\text{POC}}^*$  was moderately negatively skewed (skewness =  $-0.91$ ) and  $F_Q^*$  was slightly positively skewed (skewness =  $0.40$ ), while the distribution of  $w'$  is nearly symmetric around 0 (skewness =  $0.10$ ; Figure S7 in Supporting Information S1).

$F_{\text{POC}}^*$  and  $F_Q^*$  were spatially heterogeneous (Figure 4; Figure S13 in Supporting Information S1). Upstream ( $x < 84 \text{ km}$ ), regions of enhanced downward  $F_{\text{POC}}^*$  were clustered along the dense side of the front. Spatial distributions upstream point to the contribution of along-isopycnal transport associated with frontal convergence and downwelling. Here,  $F_{\text{POC}}^*$  reached up to  $\sim -1,450 \text{ mg C m}^{-2} \text{ d}^{-1}$  downward and  $F_Q^*$  reached up to  $\sim 5,500 \text{ W m}^{-2}$  in a region of enhanced positive vorticity and strain along the dense side of the front ( $\zeta > f$ ; Figures S3c, S3e, S3h, S3j, S3m, and S3o in Supporting Information S1).  $\delta H' \text{POC}'$  and  $\delta H' Q'$ , representing instantaneous advective fluxes due to mixed layer restratification, were about an order of magnitude smaller than  $F_{\text{POC}}^*$  and  $F_Q^*$  in the upstream region (Figure S14 in Supporting Information S1). The averages of  $\delta H' \text{POC}'$  and  $\delta H' Q'$  ( $14.7 \text{ mg C m}^{-2} \text{ d}^{-1}$  (upward),  $-32.3 \text{ W m}^{-2}$  (downward)) were less than half  $F_{\text{POC}}$  and  $F_Q$  in this region ( $-49.0 \text{ mg C m}^{-2} \text{ d}^{-1}$  (downward),  $86.6 \text{ W m}^{-2}$  (upward)).

Downstream ( $x > 84 \text{ km}$ ), downward  $F_{\text{POC}}^*$  were associated with submesoscale filaments. For example, the enhancement of  $F_{\text{POC}}^*$  in SF1 was due to co-locations of positive POC anomalies and strong convergences leading to subduction (Figures S15c–S15d, S15h, and S15i in Supporting Information S1). SF1 was also associated strengthening positive vorticity and strain between Oct. 26M and Oct. 26A (Figures S15a, S15b, S15f, and S15g in Supporting Information S1). Here, values of  $F_{\text{POC}}^*$  reached  $\sim 1,100 \text{ mg C m}^{-2} \text{ d}^{-1}$  downward, which was associated with a weaker maximum  $F_Q^*$  of  $650 \text{ W m}^{-2}$ .

Spatial distributions of  $\text{POC}'$ ,  $\text{SST}'$ , vertical velocities,  $F_{\text{POC}}^*$ , and  $F_Q^*$  were assessed as a function of distance from the front to further illuminate the regions most significantly contributing to net POC and heat fluxes. Data was binned into 5 km bins, but variability across the front was not sensitive to bin size (results using 1 and 3 km bins plotted in Figure S16 in Supporting Information S1). Only areas with coverage by all 3 overpasses were included in the analysis.  $\text{POC}'$  is negative (positive) on the light (dense) side of the front within 10 km from the front (Figure 5a).  $\text{SST}'$  exhibits the opposite pattern (Figure 5a). Vertical velocity anomalies ( $w'$ ) experienced weaker trends as a function of distance to the front and remain close to 0 when averaged in space. Mean  $w'$  is slightly negative at  $x = 5 \text{ km}$  in the upstream domain likely due to ageostrophic secondary circulation.

Negative mean  $F_{\text{POC}}^*$  and positive mean  $F_Q^*$  are elevated on the dense side of the front at  $x = 5 \text{ km}$  (mean  $F_{\text{POC}}^*$ :  $-88.3 \text{ mg C m}^{-2} \text{ d}^{-1}$  and mean  $F_Q^*$ :  $415.3 \text{ W m}^{-2}$ ). Mean  $F_{\text{POC}}^*$  and  $F_Q^*$  at  $x = 5 \text{ km}$  have large standard deviations, suggesting significant along-front variability. We also conducted the analysis on the upstream ( $x < 84 \text{ km}$ ) and downstream ( $x > 84 \text{ km}$ ) domains separately to isolate their contributions. Mean  $F_{\text{POC}}^*$  and  $F_Q^*$  at  $x = 5 \text{ km}$  are more pronounced when only the upstream domain of the front is considered, suggesting the influence and contribution of frontal overturning and subduction on the dense side of the front to net fluxes (mean  $F_{\text{POC}}^*$ :  $-182.4 \text{ mg C m}^{-2} \text{ d}^{-1}$ ; mean  $F_Q^*$ :  $587.9 \text{ W m}^{-2}$ ; thin solid black lines in Figures 5c and 5d). Downstream, the largest negative mean  $F_{\text{POC}}^*$  occurs at  $x = -5 \text{ km}$  on the light side of the front, where submesoscale filaments deviate from the smooth frontal boundary ( $-123.8 \text{ mg C m}^{-2} \text{ d}^{-1}$ ; Figure 5c).



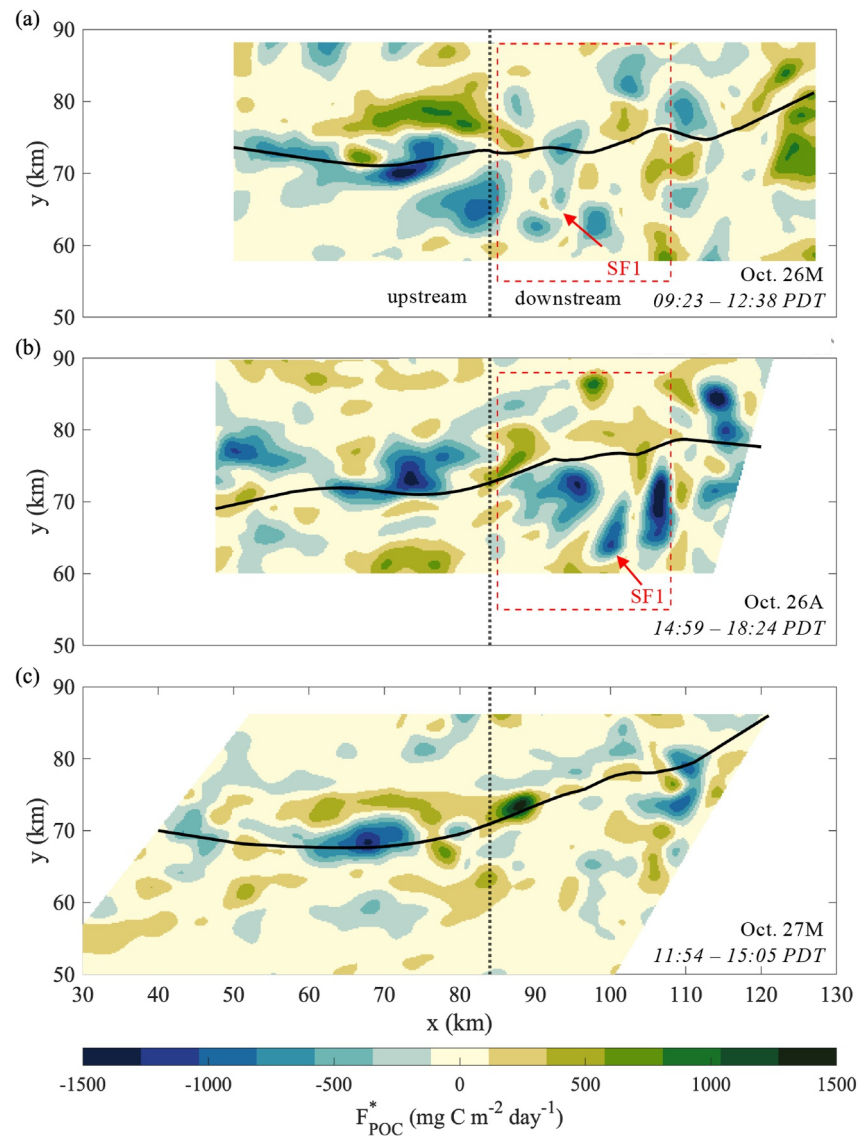
**Figure 3.** 2D histogram of  $F_Q^*$  plotted against  $F_{POC}^*$ . Linear fit denoted by black dashed line in black. Overlapping histograms of  $F_{POC}^*$  (green) and  $F_Q^*$  (pink) (b).

#### 4.2. Vertical Frontal Structure

We examined the hydrographic and bio-optical properties of the front to assess the vertical extent of subduction and characterize the conditions impacting advective fluxes. Evidence of along-isopycnal transport is captured in the averaged mesoscale front transect as salty, high chl-fl and high POC waters extending from the surface downward along the 24.5 kg/m<sup>3</sup> isopycnal (Figures 6a–6c). These profiles represent the vertical structure of frontal overturning in the upstream domain. Density and buoyancy gradients are driven by temperature above the 25 kg/m<sup>3</sup> isopycnal (Figure S17 in Supporting Information S1). Hence, salinity acts similarly to a passive tracer and is used here to trace subducted waters. The high POC intrusions are transported below the mixed layer, but do not penetrate the pycnocline. Along the 24.5 kg m<sup>-3</sup> isopycnal and under the light side of the front ( $x = -10$  to  $-15$  km), there is a shift from high chl-fl and high POC waters to low chl-fl and high POC waters. The change in bio-optical properties may indicate a shift to different phytoplankton communities or physiological changes (Figures 6a and 6b). The high chl-a and high POC subducted water slides above a deep photo-acclimation maximum (DAM) between the 24.5 and 25 kg/m<sup>3</sup> isopycnals, which is undetected by remote sensing and marked by high chl-fl but low POC and biomass (Cornec et al., 2021; Figures 6a and 6b).

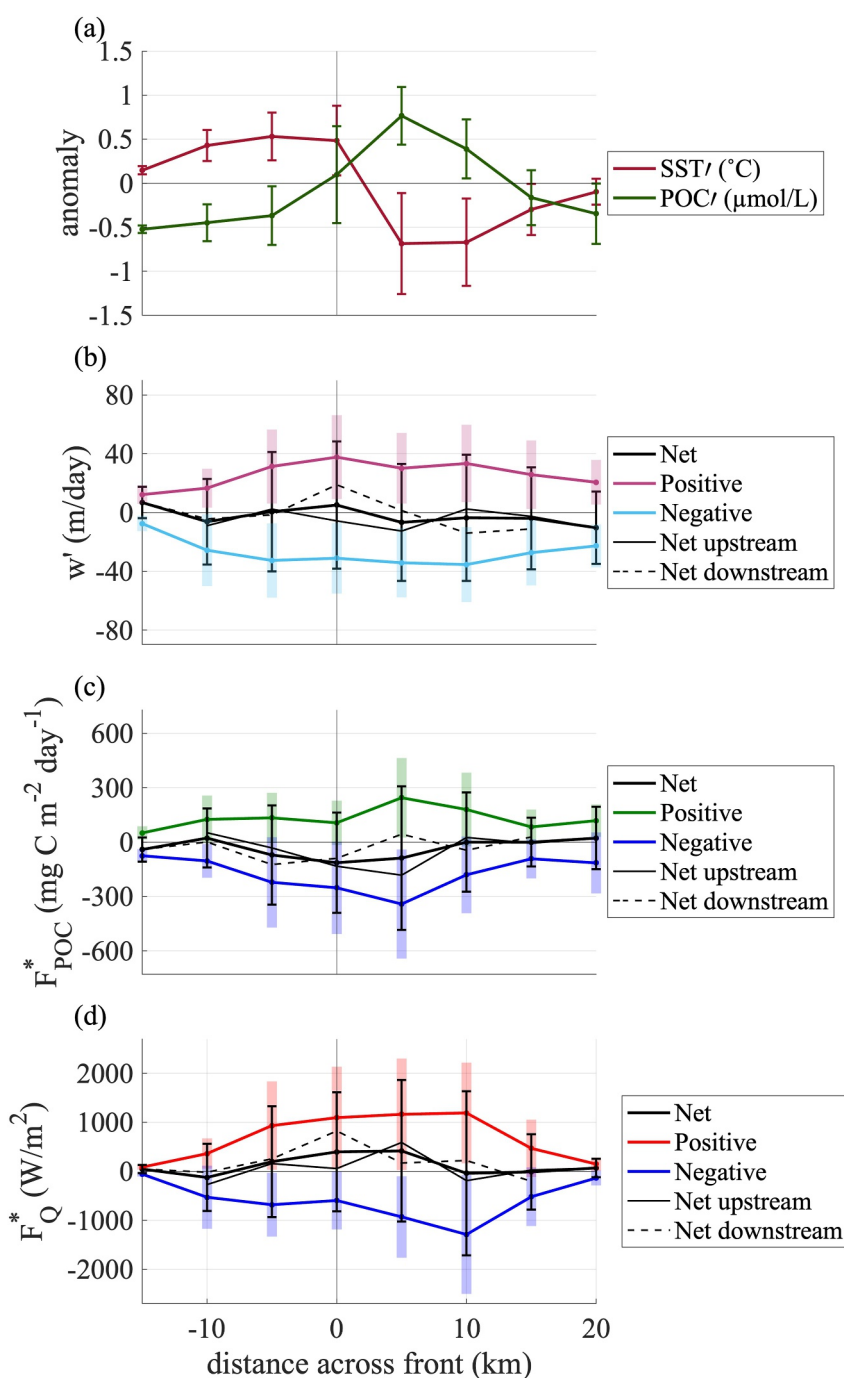
Strong lateral buoyancy gradients at the front ( $M^2 \approx -1.0 \times 10^{-6} \text{ s}^{-2}$ ) and slumping isopycnals suggest that the front was restratifying at the time of data collection (Figure 6e). The front was also marked by increased stratification ( $N^2$ ) associated with isopycnal outcropping (Figure 6f).  $N^2$  was weak on both sides of the front and elevated along the 24.5 kg m<sup>-3</sup> isopycnal which represented the base of the subduction (Figure 6f). Average mixed layer varied systematically across the front. The mixed layer depth on the less dense side of the front ( $x = -25$  to  $-15$  km) ranges from  $H = 25$ – $32$  m until it began to shoal approaching the front (between  $x = -15$  to  $0$  km). Then,  $H$  began to rapidly deepen on the dense side of the front ( $H \approx 32$  m between  $x = 0$ – $5$  km) in a region of low stratification. Away from the front ( $x > 5$  km), the mixed layer depth begins to shoal again ( $H = 15$ – $22$  m at  $x > 12$  km). Flow was oriented approximately along the front, and zonal velocities were enhanced around the front down to  $\sim 100$  m depth. (Figure S18a in Supporting Information S1). Meridional velocities across the front ( $v$ ) were weak but oriented toward the front on both sides, indicating a surface convergence at the front (Figure S18b in Supporting Information S1).

The vertical advection of SF1 is captured in the submesoscale filament transect as a tongue of salty (33–33.1 g/kg), high chl-fl, high POC water which slid below the mixed layer ( $x = 0$ – $10$  km between the 24 and 24.5 kg/m<sup>3</sup> isopycnals; Figures 6g–6i). This profile represents the vertical structure of filament subduction in the downstream domain. The vertical profiles indicated that SF1 had been subducted underneath the shallow mixed layer ( $H \approx 15$  m). Convergent flow at the filament, estimated by DopplerScatt (Figure S15 in Supporting Information S1), may have brought relatively oligotrophic and fresh (32.8–32.9 g/kg) surrounding waters toward



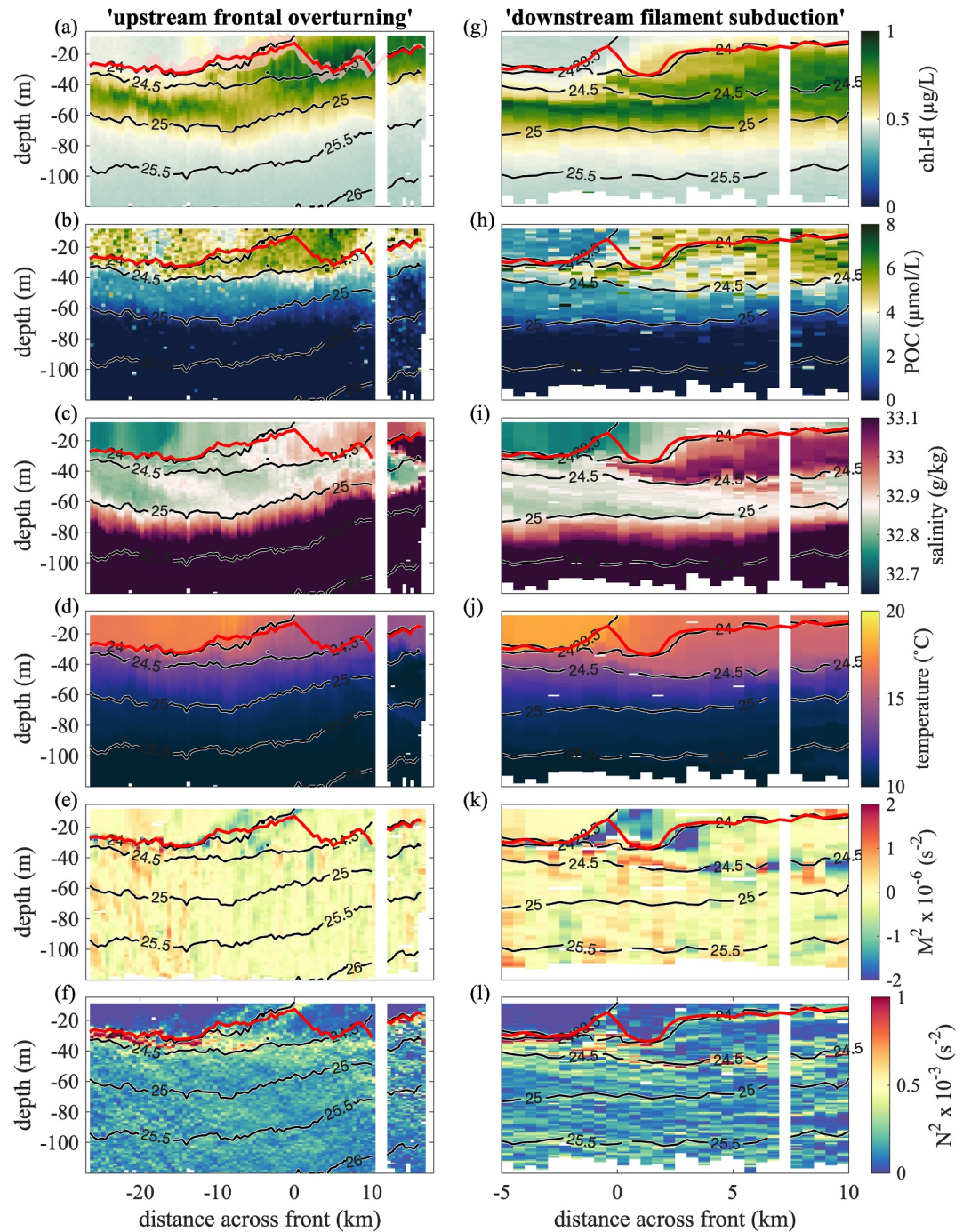
**Figure 4.** Maps of estimated  $F_{POC}^*$  for the three remote sensing snapshots Oct. 26M (a), Oct. 26A (b), and Oct. 27M (c). Green indicates positive, or upward, fluxes of particulate organic carbon (POC) caused by either the upwelling of high POC water or downwelling of low POC water. Blue indicates negative, or downward, fluxes of POC caused by either the downwelling of high POC water or the upwelling of low POC water. SF1 represents subducted filament of interest plotted in Figure S15 in Supporting Information S1 (area plotted in figure denoted by dashed red lines). Smoothed sea surface temperature front shown by solid black line. Dashed vertical line represents upstream and downstream demarcations.

the filament, which slid underneath the lighter waters. Surface water above the subducted filament is fresher than the filament but saltier than the offshore water, suggesting the presence of mixing (Figure 6i).  $M^2$  reached higher values at the edge of the filament than at the mesoscale front ( $M^2 \approx -2.5 \times 10^{-6} \text{ s}^{-2}$ ; Figure 6k). Elevated  $N^2$  along the  $24.5 \text{ kg m}^{-3}$  isopycnal represents the base of filament subduction. As at the mesoscale front, the mixed layer depth is highly variable at SF1 ( $H = 15\text{--}34 \text{ m}$ ) and rapidly deepens within the dense filament ( $x = 0\text{--}2.5 \text{ km}$ ; Figures 6g–6l). This region of deepened mixed layer is also marked by enhanced zonal flow at the surface down to the base of the mixed layer ( $x = 0\text{--}2.5 \text{ km}$ ; Figure S18c in Supporting Information S1).



**Figure 5.** Five-kilometer binned and averaged sea surface temperature (red) and particulate organic carbon (green) anomalies plotted against distance across front (a). Binned and averaged positive (pink), negative (light blue), and net (black) vertical velocities plotted against distance across front (b). Binned and averaged positive (green), negative (blue), and mean  $F_{\text{POC}}^*$  (black) plotted against distance across front (c). Binned and averaged positive (red), negative (blue), and mean  $F_Q^*$  (black) plotted against distance across front (d). Thin solid black lines represent net averages in the upstream domain only, and thin dashed black lines represent net averages downstream (b–d). Error bars represent  $\pm 1$  standard deviation away from the mean.





**Figure 6.** Vertical EcoCTD profiles of hydrographic and bio-optical properties, averaged and binned (500 m horizontal bins) in an across frontal reference frame. Chlorophyll-fluorescence (chl-fl, a, g), particulate organic carbon (POC, b, h), salinity (c, i), temperature (d, j),  $M^2$  (e, k), and  $N^2$  (f, l) plotted against distance across front on the x-axis. The average mesoscale front transect represents the vertical structure of upstream frontal overturning (a–f) and the submesoscale filament transect surveying SF1 represents the vertical structure of downstream filament subduction (g–l). Isopycnals overlain in black and labeled with corresponding density anomalies. Averaged mixed layer depth plotted in red (a–l). Standard deviation plotted in shaded pink (a).

## 5. Discussion

### 5.1. Connections Between (Sub)mesoscale Dynamics, Vertical Heat Fluxes, and Submesoscale-Driven Carbon Export in Upwelling Systems

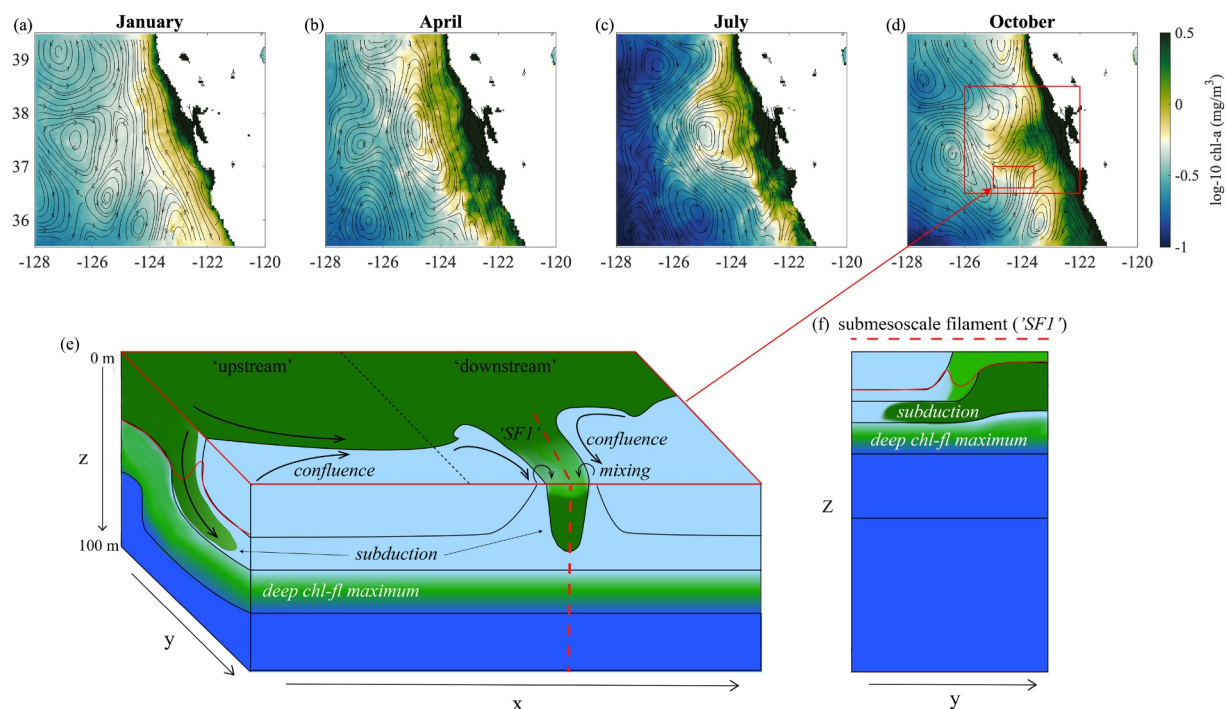
In this manuscript, we describe the first application of high resolution, DopplerScatt derived vertical velocities combined with satellite ocean color to directly estimate the net physical flux of POC at a front. We found a net vertical downward POC flux and net upward heat flux at a restratifying front in the California Current System. Frontal overturning and instabilities at fronts are associated with positive buoyancy fluxes that act to restratify the mixed layer (Capet et al., 2008a). Buoyancy gradients in the CCS are typically set up by interactions between cold, upwelled waters and warmer offshore waters, so therefore a positive buoyancy flux typically leads to a positive vertical heat flux. In our study, we show that submesoscale features at the front are also associated with downward POC flux, as temperature and POC are anti-correlated due to the enhancement of nutrients and phytoplankton in cold upwelled waters. The dynamics and transport associated with the front highlighted in this study are likely representative of other upwelling fronts in the CCS. Johnson et al. surveyed the evolution of a similar upwelling front offshore of San Francisco, characterized by  $M^2$  of  $\mathcal{O}(10^{-6} \text{ s}^{-2})$ , shallow mixed layers ( $\sim 30 \text{ m}$ ), and vorticity, strain, and divergence of  $\mathcal{O}(f)$  (L. Johnson, Lee, D'Asaro, Thomas, & Shcherbina, 2020; L. Johnson, Lee, D'Asaro, Wenegrat, & Thomas, 2020). The study found that sharp fronts in shallow mixed layers can be associated with significant kinetic energy and buoyancy fluxes occurring on rapid (super-inertial) time scales, which can have implications for momentum and heat budgets (L. Johnson, Lee, D'Asaro, Thomas, & Shcherbina, 2020; L. Johnson, Lee, D'Asaro, Wenegrat, & Thomas, 2020). Our results show that these types of rapidly restratifying fronts can also make significant contributions to POC flux.

Submesoscale fronts and instabilities can contribute disproportionately to advective fluxes of carbon, highlighting the importance of resolving the submesoscale to accurately diagnose the contributions of fronts in the CCS to carbon export (Ruiz et al., 2019). Mesoscale fronts host large downward fluxes of nitrate and total organic carbon, in addition to POC, in the CCS (Nagai et al., 2015). Stukel et al. (2017) estimated that mesoscale fronts may account for over 25% of total carbon sequestration in the CCS and other Eastern Boundary Upwelling Systems due to subduction and the enhancement of gravitational sinking (2019). Assessing net flux estimates at varying spatial averaging scales shows that net flux magnitudes are significantly decreased when assessed at coarser resolutions. The analysis indicates that resolving submesoscale-driven, instantaneous advective fluxes are important for quantifying net estimates of physical POC export at fronts (Figure S9 in Supporting Information S1). In this study, submesoscale instabilities were likely evident through the meanders and filaments shown in the chl-a, SST, vorticity, and strain fields where conditions were favorable for the development of symmetric and mixed layer instabilities (Text S4 in Supporting Information S1). Submesoscale instabilities, such as symmetric instabilities (SI) and mixed layer instabilities (MLI), can generate significant vertical velocities and fluxes. More work is needed to determine how scale resolution impacts POC export estimates over larger spatiotemporal domains.

### 5.2. Seasonal and Geographic Influences on Submesoscale-Driven Subduction

Seasonal changes in submesoscale vertical velocities, POC distributions, and their drivers in Eastern Boundary Current Upwelling Systems (EBUSs) may impact the contribution of submesoscale-driven flux to total POC export on annual timescales. Wind-driven coastal upwelling varies seasonally in all EBUSs and is linked to frontal activity (Castelao et al., 2006) and mesoscale stirring intensity in the California Current System (Giddings et al., 2022). Coastal upwelling is also the main mechanism of nitrate supply for primary production, modulating the large-scale seasonal variability in phytoplankton and POC distributions (García-Reyes & Largier, 2012; Messié & Chavez, 2015; Messié et al., 2023). Upwelling off central and northern California is the strongest from April to June, yielding an increase in chl-a concentrations from March/April to maximum values in June (García-Reyes & Largier, 2012). Drivers of primary productivity beyond nutrient limitation, which includes physical transport by subduction and lateral flow, iron supply, and light limitation, also vary seasonally (Messié & Chavez, 2015). Changes in seasonal drivers impact the seasonality of POC and therefore the impact of submesoscale dynamics on POC flux.

The front analyzed in this study is a part of a consistent feature of the seasonal mesoscale field. Monthly climatologies representing different seasons of chl-a fields (proportional to POC) and geostrophic streamlines, show a plume of high chl-a water being transported offshore by persistent mesoscale eddies (October, Figure 7d). The



**Figure 7.** Monthly Sentinel-3A chl-a climatologies (2018–2024; NASA Ocean Biology Processing Group) overlain by geostrophic streamlines from absolute geostrophic velocity anomalies (1993–2022; SSALTO/DUACS) for January (a), April (b), July (c), and October (d). Study region and frontal region plotted in Figure 2 denoted by red boxes in panel (d). The seasonal mesoscale distributions set the background gradients for the submesoscale processes that drive the vertical fluxes. Panels (e–f) show a three-dimensional depiction of the processes occurring at the front due to upstream frontal overturning and downstream filament subduction. In the upstream domain, vertical transport is depicted as high particulate organic carbon (POC) water (dark green) being advected downward along the isopycnal on the dense side of the front. POC is subducted below the variable mixed layer in red (e). Flow is confluent toward the front. In the downstream domain, SF1 represents a dense filament with high POC concentrations at the surface (e). SF1 forms as an instability from the front and sinks below the shallow mixed layer (f). The fresher, surrounding water converges toward the filament. Subduction occurs above the deep chl-fl maximum (light green), representing a deep layer of phytoplankton photo-acclimated to low light levels (f).

seasonality of mesoscale variability in the CCS is a well documented phenomena and impacts offshore transport of POC (Amos et al., 2019; Giddings et al., 2022; Jorgensen et al., 2024). The plume of chl-a, recurring climatologically in this latitude band, resembles the mesoscale conditions in this study (Figures 2 and 7). Therefore, it is possible that the dynamics and fluxes captured in this study are characteristic of this seasonally reoccurring feature, but more work is needed to quantify the feature's interannual variability. Additionally, the transport of the cold, high chlorophyll plume offshore means that carbon export due to submesoscale advective fluxes is associated with longer timescales than if occurring on the shelf.

The impact of submesoscale dynamics on submesoscale-driven POC flux also depends on their influences on primary productivity. Submesoscale currents can act to fuel or inhibit new production. In EBUSs such as the California Current System, fronts and eddies have been shown to reduce production in coastal waters and enhance production further offshore in more oligotrophic waters (Gruber et al., 2011; Kessouri et al., 2020). The front surveyed in this study was located in coastal productive waters, where vertical nutrient gradients are negative due to an enrichment of nutrients at the surface. Subduction, therefore, leads to a net decrease in surface nutrients, whereas offshore, subducted surface water is low in nutrients and upwelled water is relatively enriched (Kessouri et al., 2020). Additionally, vertical advection of phytoplankton through gradients of nutrients and light availability can have significant impacts on primary productivity and phytoplankton patchiness (Mahadevan, 2016). Overall, a holistic parameterization of submesoscale impacts on POC flux in models should also include seasonal drivers and the interactions between submesoscale dynamics and primary production.



### 5.3. Relationships Between Surface Kinematics and Vertical Fluxes

Linking surface measurements to vertical transport is important for understanding the dynamics leading to flux and expanding the spatiotemporal scope of vertical flux estimates. Submesoscale dynamics can be characterized by their distributions in vorticity-strain space (Balwada et al., 2021). We assessed the relationships between vorticity, strain, and vertical fluxes to examine co-variations between fluxes and dynamical regimes (Balwada et al., 2021; Shcherbina et al., 2013). Counts, distance across front, POC fluxes, and vertical heat fluxes were binned into  $0.1 \times 0.1$  vorticity-strain bins. The overall vorticity-strain diagram was skewed toward positive vorticity ( $+\zeta/f$ ), characteristic of submesoscale frontogenesis (Figure 8a; Lenain et al., 2023; Shcherbina et al., 2013). Regions close to the front along  $\sigma/f = |\zeta/f|$  hosted negative  $F_{\text{POC}}^*$  and positive  $F_Q^*$  on average, further pointing to the role of frontogenesis and restratification in driving observed vertical advective fluxes.

A hotspot of large, negative POC fluxes and positive vertical heat flux occurred close to the front and in a cyclonic-dominated dynamical region (Figures 8c and 8d). The hotspot represents the strong downward fluxes along the dense side of the front in the upstream domain shown in Figure 4. Other notable features were two hotspots of negative POC flux in the strain-dominated region far from the front, which were associated with both positive and negative vertical heat fluxes (Figures 8c and 8d). These regions are associated with submesoscale eddies in both the upstream and downstream domains. We also further assessed the conditions that led to filament subduction by studying the evolution of kinematic properties of SF1 between Oct. 26M and Oct. 26A (Text S3, Figure S15 in Supporting Information S1). Overall, the highest density of observations lied in the low vorticity - low strain region of the vorticity-strain diagram and are associated with small advective fluxes, suggesting that net flux is driven by large episodic advective fluxes in regions with high vorticity or strain.

$F_{\text{POC}}^*$  and  $F_Q^*$  were anti-correlated generally, but their relationships to surface kinematics differed.  $F_Q^*$  may have been reduced at eddies due to partial salinity compensation reducing its correlation with buoyancy. Additionally,  $F_{\text{POC}}^*$  and  $F_Q^*$  may have become misaligned due to increased productivity in nutrient-rich, cold waters, increasing POC' relative to the SST'. Expanding observational datasets linking surface kinematics with tracer fluxes could improve the understanding of relationships between surface measurements and vertical transport processes.

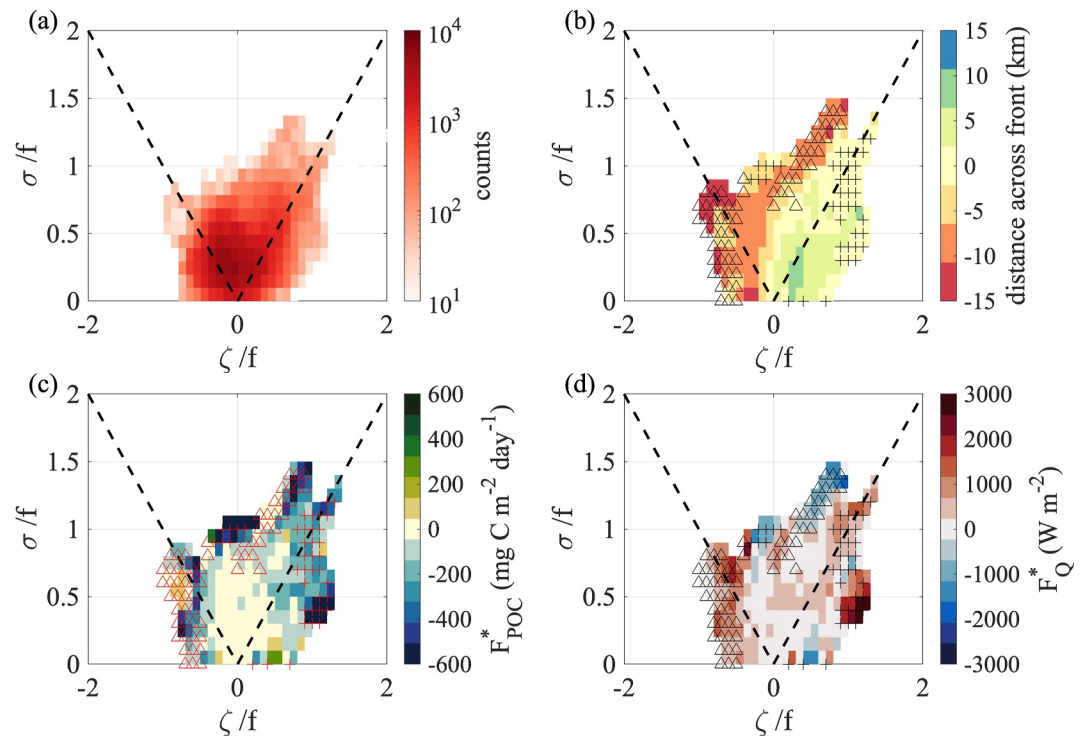
### 5.4. Limitations and Uncertainties

We assumed that the vertical scale of the submesoscale fluxes scaled with the mixed layer depth. Further characterization of the vertical scale and dynamics driving flux through focused modeling and theoretical work could help improve flux estimates. We also assumed that divergence is constant with depth over the mixed layer. However, the true vertical scale is difficult to quantify due to the difficulty of observationally resolving the 3D structure of divergence. We assessed the vertical structure of divergence using 3D velocity fields from Regional Ocean Modeling System (ROMS) output of the S-MODE study region (Text S1 in Supporting Information S1). Divergence experienced low variability within the mixed layer relative to horizontal variability in divergence (Figure S4 in Supporting Information S1), but more work is needed to assess how vertical variability in horizontal divergence impacts flux estimates.

Overall, assessing the approach presented in this study in other oceanic regions is critical before implementation and interpretation. In a Reynolds decomposition, flux has two eddy flux components; the first component, reported in this study, is the flux due to the covariance of vertical velocities and POC ( $Q$ ) anomalies (Text S2 in Supporting Information S1). The second component, explored in the upstream domain, represents the flux due to mixed layer restratification ( $\overline{\delta'H'POC'}$ ;  $\overline{\delta'H'Q'}$ ). Overturning circulation at the front led to an enhanced downward transport of POC along sloping isopycnals on the dense side of the front (Figure 4). However, the spatial patterns of the mixed layer restratification flux were in the opposite direction at the front and overall less than half of  $\overline{\delta'H'POC'}$ . Mixed layer shoaling at the front decreased the extent of vertical mixing which led to upward  $\overline{\delta'H'POC'}$  (Figure S14 in Supporting Information S1). Estimates of the flux due to mixed layer restratification are less robust than  $\overline{\delta'H'POC'}$  due to smaller spatiotemporal averaging extents and uncertainty introduced by the mixed layer anomaly term. Future work should attempt to estimate the relative contribution of mixed layer depth variability to flux, which will vary by oceanic region.

The Reynolds decomposition approach to submesoscale vertical flux estimation requires sufficient data for extended averaging. However, even with extended averaging, net flux, or turbulent flux, estimates tend to show





**Figure 8.** Joint probability distribution function of vorticity and strain (a; Shcherbina et al., 2013). Distance from front (b),  $F_{\text{POC}}^*$  (c), and  $F_Q^*$  (d) plotted in vorticity-strain space. Dashed lines represent 1:1 lines. Cross-hatching and triangles indicate areas where  $>80\%$  of the data points are from the upstream or downstream domains (black in panels (b, d); red in panel (c)). Each bin has a minimum number of 10 data points.

poor convergence to their true values, because the distribution of instantaneous advective fluxes is typically highly intermittent and heavy-tailed. Rare but intense events, such as submesoscale frontal instabilities or convective plumes, can dominate the flux variance, causing the estimated instantaneous advective fluxes  $w' \text{POC}'$  and  $w' Q'$  to fluctuate strongly. Decreasing net flux magnitudes at coarser spatial resolutions points to the role of small-scale, episodic events in driving vertical transport (Figure S9 in Supporting Information S1). Hence, net flux estimation is sensitive to the spatiotemporal scope of averaging, of which airborne snapshots are limited.

To quantify the net contribution of submesoscale advective fluxes to POC export in the CCS, fluxes must be integrated over much larger spatiotemporal scales than at a single front. In this study, we estimated the submesoscale surface flux through the mixed layer, associated with snapshots of a single restratifying upwelling front. The vertical profiles revealed that subduction occurred below the mixed layer, but not the pycnocline, where long-term isolation from the atmosphere occurs (Figure 6). If subducted POC is not re-entrained by upward vertical velocities and remains below the mixed layer, it has a higher chance of eventual sequestration on climate-relevant timescales. Surface fluxes indicate how much carbon has the potential for export, but quantifying the percentage of carbon that is eventually sequestered would require integrating over longer time scales such as with a high-resolution process model. Lastly, integrating a hyperspectral ocean color instrument on the B-200 for concurrent measurements with DopplerScatt could increase the accuracy of flux estimates at these submesoscale features to eliminate temporal separation between measurements.

## 6. Summary

In this study, we demonstrated the first application of combining high resolution vertical velocities from airborne scatterometry (NASA DopplerScatt) with satellite ocean color to directly estimate the net physical flux of POC at an upwelling front in the California Current System. Instantaneous, advective fluxes of POC and heat were anti-correlated ( $R^2 = 46\%$ ) and reached values up to  $\mathcal{O}(1,000) \text{ mg C m}^{-2} \text{ day}^{-1}$  and  $\mathcal{O}(1,000) \text{ W m}^{-2}$  both upward and downward. When integrated over sufficient spatiotemporal scales to achieve a negligible total volume flux, the

turbulent flux, or net flux, can be estimated. Overall, we found that submesoscale dynamics made important contributions to physical POC export ( $44.1 \pm 23.1 \text{ mg C m}^{-2} \text{ day}^{-1}$  downward) and vertical heat flux ( $186.2 \pm 101.6 \text{ W/m}^2$  upward) at a seasonally re-occurring upwelling front in the California Current System. We characterized two dynamical regimes at the front. In the first upstream domain, representing an earlier stage of frontal evolution, instantaneous downward advective fluxes were concentrated along the dense side of the front in a region of enhanced vorticity and strain, pointing to the role of frontal overturning. Here, POC subducted along-isopycnals and below the mixed layer was captured in a series of high resolution vertical profiles across the front. In the second downstream domain, representing a later, unstable regime, downward advective fluxes were associated with filaments that disappeared within a couple of days. This study suggests that parameterizing submesoscale-driven POC flux due to frontogenesis and instabilities in models is important for constraining physical carbon export in highly-productive Eastern Boundary Upwelling Systems. This measurement approach can be expanded to other oceanic regions to further understand the relationships between vertical heat and carbon fluxes, and ultimately, the impact of submesoscale vertical velocities on the marine carbon cycle.

## Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

## Availability Statement

Data in this paper was collected during the Submesoscale Ocean Dynamics Experiment (S-MODE). The S-MODE data presented in this manuscript is publicly available via the Physical Oceanography Distributed Active Archive Center (PO.DAAC) at <https://podaac.jpl.nasa.gov/S-MODE>. S-MODE in situ data sets used in this manuscript were S-MODE L2 Shipboard Acoustic Doppler Current Profiler Measurements Version 1 (<https://doi.org/10.5067/SMODE-RVADC>; S-MODE Team, 2022a), S-MODE L2 Shipboard Thermosalinograph, Meteorology, and Bio-optics Measurements Version 1 (<https://doi.org/10.5067/SMODE-RVTSG>; S-MODE Team, 2022b), S-MODE L2 Shipboard Bottle Data Version 1 (<https://doi.org/10.5067/SMODE-RVBOT>; Lang et al., 2022), S-MODE Shipboard uCTD and EcoCTD Measurements Version 1 (<https://doi.org/10.5067/SMODE-RVECT>; S-MODE Team, 2022c), and S-MODE L4 Regional Ocean Modeling System (ROMS) Model Output Version 1 (<https://doi.org/10.5067/SMODE-ROMS>; Hypolite, 2025). A full description of in situ data collection and bio-optical calibrations can be found here, in the S-MODE Bio-optical Data Report for IOP-1: [https://archive.podaac.earthdata.nasa.gov/podaac-ops-cumulus-docs/s-mode/docs/S-MODE\\_IOP1\\_2022\\_BIO\\_OPTICS\\_DATA\\_REPORT.pdf](https://archive.podaac.earthdata.nasa.gov/podaac-ops-cumulus-docs/s-mode/docs/S-MODE_IOP1_2022_BIO_OPTICS_DATA_REPORT.pdf) (Lang et al. (2023)). S-MODE MOSES Level 2 Atmospherically-Corrected SST Version 1 can be found at <https://doi.org/10.5067/SMODE-MOSE2> (Molemaker & Hypolite, 2022). S-MODE DopplerScatt Level 2 Ocean Winds and Currents Version 2 can be found at <https://doi.org/10.5067/SMODE-DSCT2-V2> (Rodriguez et al. (2023)). Sentinel-3A and 3B SLSTR Level-2 Full Resolution Water Surface Temperature (WST), as well as OLCI Level-2 Full Resolution chlorophyll-a (WFR, Water Full Resolution), are available through the European Space Agency's Copernicus Browser ([browser.dataspace.copernicus.eu](https://browser.dataspace.copernicus.eu)). Sentinel-3 chlorophyll-a monthly climatologies, used in Figure 7, were collected with the NASA Level 3 and 4 Ocean Color Browser from the Ocean Biology Processing Group (OBPG) through the Ocean Biology DAAC (OB.DAAC) (<https://oceandata.sci.gsfc.nasa.gov/l3/>). Gridded monthly climatology maps of sea level anomaly geostrophic velocities, used in Figure 7, were processed by SSALTO/DUACS (Developing Use of Altimetry for Climate Studies) and distributed by AVISO+ (<https://www.aviso.altimetry.fr>) with support from CNES (<https://www.aviso.altimetry.fr/index.php?id=1526>). Global Total (COPERNICUS-GLOBCURRENT) Ekman and Geostrophic currents at the Surface and 15 m, used in Figures 2a–2d, from Copernicus Marine Environment Monitoring Service (Rio et al., 2014).

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